MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED LOW POWER SCHOTTKY TTL, NAND GATES, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/370(USAF), dated 18 March 1983, and is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Delete and substitute as printed above.

* 1.2.3, case outline, outline letter: Delete "X" and substitute "2". Delete "Y".

1.2.3, case outline: Delete "square chip carrier, option A" and substitute "square chip carrier package".

1.2.3, case outline: Delete "C-2 (20-terminal, .350" x .350"), square chip carrier, option B", in its entirety.

PAGE 2

1.3, thermal resistance, delete and substitute:

"Thermal resistance, junction-to-case (Θ_{JC}):
Cases A, B, C, D, E, F, and 2 - - - - - - - - - - See MIL-M-38510, appendix C.

1.3, junction temperature (T_J): Add "2/".

1.4, supply voltage: Add "(V_{CC})".

3.2.4, schematic circuits, delete and substitute:

"3.2.4 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request."

Bottom of page, add the following footnote:

"2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening condition in accordance with method 5004 of MIL-STD-883."

PAGE 3

* Table I, output current, I_O, device types 02, 04, and 05, min limits column: Delete "-30" and substitute "-20".

Table I, I_{IL}, max limits column: Delete "-200" and substitute "-400".

Table I, I_{CCL}, max limits column, device type 06: Delete "3.8" and substitute "4.2".

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PAGE 4

Table I, t_{PHL}, limits column, device type 02: Delete "4" and "21" and substitute "3" and "12", respectively.

* Table I, t_{PHL}, max limits column, device type 03: Delete "12" and substitute "15" (alters change made by amendment 2).

Table I, t_{PLH}, max limits column, device type 02: Delete "14" and substitute "15".

Table I, t_{PLH}, max limits column, device type 06: Delete "14" and substitute "17".

PAGE 5

4.2a(1): Add "or E" as deleted in amendment 1.

4.2a(1): Add burn-in condition F.

4.2c, delete in its entirety and substitute:

"c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510".

4.4.2, group B inspection, add the following new sentence: "Electrical parameters shall be as specified in table II herein".

PAGE 6

4.4.3c(1): Add "or E" as deleted in amendment 1.

4.4.3c(1): Add test condition F.

6.1, notes, delete in its entirety.

PAGE 7

Figure 1: Delete "Cases X, Y" and substitute "Case 2" two places. Makes these same changes on pages 8 and 9, two places each page.

PAGE 12

Figure 3, device type 01, note 2: Delete "R1 = 1 kΩ ±5%;" note 3, delete "V_{CC} shall be high enough" and substitute "V_{CC} and R1 shall be chosen".

Figure 3, device type 02, note 2: Delete "R1 = 10Ω ±5%;" note 3, delete "V_{CC} shall be high enough and substitute "V_{CC} and R1 shall be chosen". Makes these same changes on page 13, notes 2 and 3.

PAGE 14

Figure 3, note 2: Delete "R1 = 10Ω maximum;".

Figure 3, note 3: Delete "V_{CC} shall be high enough" and substitute "V_{CC} and R1 shall be chosen".

PAGE 15

Figure 4, note 1: Delete "PRR = 1.0 MHz" and substitute "PRR ≤ 1.0 MHz".

PAGE 16

Table III: Delete "Cases X, Y" and substitute "Case 2". Make this same change on pages 17 through 27.

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PAGE 17

Table III, footnote 1/, delete "cases X and Y" and substitute "case 2". Make this same change on pages 19, 21, 23, 25, and 27.

Table III, footnote 2/, circuit C limit: Delete "0/-100" and substitute "-0.2/-400".

PAGE 18

Table III, test 42, pin 1C: Delete "5.5 V" and substitute "0.4 V".

Table III, test 43, pin 1C: Delete " " and substitute "5.5 V".

PAGE 19

Table III, test numbers 54 through 62, min, max limits columns: Delete "4" and "18" and substitute "3" and "11", respectively.

Table III, test numbers 63 through 71, max limits column: Delete "11" and substitute "12".

Table III, subgroup 10, t_{PHL}, min, max limits column: Delete "4" and "21" and substitute "3" and "12", respectively.

Table III, subgroup 10, t_{PLH}, max limits column, delete "14" and substitute "15".

Table III, footnote 2/, circuit C limit: Delete "0/-100" and substitute "-0.2/-400".

* Table III, footnote 4/: Delete in its entirety and substitute "4/ I_O limits shall be -20/-112 mA for circuit A and -30/-110 mA for circuit C".

PAGE 21

* Table III, subgroup 10, t_{PHL}, max limit column: Delete "12" and substitute "15" (alters change made by amendment 2).

Table III, footnote 2/, circuit C limit: Delete "0/-100" and substitute "-0.2/-400".

PAGE 23

Table III, footnote 2/, circuit C limit: Delete "0/-100" and substitute "-0.2/-400".

* Table III, footnote 4/: Delete in its entirety and substitute "4/ I_O limits shall be -20/-112 mA for circuit A and -30/-110 mA for circuit C".

PAGE 24

Table III, test 48 pin H, test 49 pin I, test 50 pin J, test 51 pin K, test 52 pin L, test 53 pin H: Delete "0.7 V" and substitute "7.0 V" each place.

PAGE 25

Table III, footnote 2/, circuit C limit: Delete "0/-100" and substitute "-0.2/-400".

* Table III, footnote 4/: Delete in its entirety and substitute "4/ I_O limits shall be -20/-112 mA for circuit A and -30/-110 mA for circuit C".

PAGE 26

Table III, I_{CCL}, test 44, max limits column: Delete "3.8" and substitute "4.2".

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PAGE 27

Table III, subgroup 10, t_{PLH}, max limits column: Delete "14" and substitute "17".

Table III, footnote 2/, circuit C limit: Delete "0/-100" and substitute "-0.2/-400".

PAGE 28

6.6, generic-industry type: Delete "54ALS00" and substitute "54ALS00A". Delete "54ALS20" and substitute "54ALS20A". Delete "54ALS04" and substitute "54ALS04A".

NOTE: The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespectively of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Air Force - 17

Review activities:
Air Force - 19, 85, 99
DLA - ES

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-F723)

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